EXPERIMENT- 9

Simulation of Telemetry Data Ingestion to Time-Series Database and Visualization using Grafana, Power BI and Tableau

CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files
from IPython.display import display
```

1. Generate telemetry data

```
np.random.seed(42)

n = 100

timestamps = pd.date_range(
start="2026-10-29 08:00:00",
periods=n,
freq="min"
)

data = pd.DataFrame({
"timestamp": timestamps,
"device_id": np.random.choice(
["Device_01", "Device_02", "Device_03"], n
),
"temperature": np.random.normal(28, 3, n).round(2),
"humidity": np.random.normal(60, 8, n).clip(20, 95).round(2),
"battery": np.random.uniform(50, 100, n).round(2)
})

data["status"] = np.where(data["battery"] < 60, "Warning", "Normal")

print("=" * 60)
print("IoE TELEMETRY DATA INGESTION AND VISUALIZATION")
print("=" * 60)

print("\n1. TELEMETRY DATA")
print("Total records:", len(data))
print("Number of devices:", data["device_id"].nunique())
display(data.head(10))
```

2. Simulate time-series database ingestion using CSV

```
data.to_csv("telemetry_database.csv", index=False)
database = pd.read_csv("telemetry_database.csv")
database["timestamp"] = pd.to_datetime(database["timestamp"])

print("\n2. DATA INGESTION")
print("Telemetry data stored successfully.")
print("Records stored:", len(database))
display(database.head())
```

3. Statistical analysis

```
print("\n3. STATISTICAL ANALYSIS")

summary = database[
["temperature", "humidity", "battery"]
].agg(["mean", "min", "max"]).round(2)

display(summary)

print("\nDevice Status Summary")
status_counts = database["status"].value_counts().reindex(
["Normal", "Warning"], fill_value=0
)
display(status_counts.to_frame("Number of Readings"))

print("\nDevice-Wise Analytics")
device_summary = database.groupby("device_id").agg(
Average_Temperature=("temperature", "mean"),
Average_Humidity=("humidity", "mean"),
Average_Battery=("battery", "mean"),
Total_Readings=("device_id", "count")
).round(2)

display(device_summary)
```

4. Historical time-series visualization

```
plt.figure(figsize=(13, 5))
plt.plot(database["timestamp"], database["temperature"],
label="Temperature (°C)")
plt.plot(database["timestamp"], database["humidity"],
label="Humidity (%)")
plt.title("Historical Telemetry Data Visualization")
plt.xlabel("Timestamp")
plt.ylabel("Sensor Reading")
plt.legend()
```

```
plt.grid(True)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

5. Separate sensor monitoring graphs

```
fig, axes = plt.subplots(3, 1, figsize=(13, 10))

axes[0].plot(database["timestamp"], database["temperature"])
axes[0].set_title("Temperature Monitoring")
axes[0].set_ylabel("Temperature (°C)")
axes[0].grid(True)

axes[1].plot(database["timestamp"], database["humidity"])
axes[1].set_title("Humidity Monitoring")
axes[1].set_ylabel("Humidity (%)")
axes[1].grid(True)

axes[2].plot(database["timestamp"], database["battery"])
axes[2].set_title("Battery Level Monitoring")
axes[2].set_ylabel("Battery (%)")
axes[2].set_xlabel("Timestamp")
axes[2].grid(True)

plt.tight_layout()
plt.show()
```

6. Device status comparison

```
plt.figure(figsize=(7, 5))
plt.bar(status_counts.index, status_counts.values)
plt.title("IoE Device Status Comparison")
plt.xlabel("Device Status")
plt.ylabel("Number of Readings")
plt.grid(axis="y", linestyle="--", alpha=0.5)
plt.tight_layout()
plt.show()
```

7. Correlation heatmap

```
correlation = database[
["temperature", "humidity", "battery"]
].corr()

plt.figure(figsize=(7, 5))
plt.imshow(correlation, cmap="coolwarm", vmin=-1, vmax=1)
plt.xticks(range(3), correlation.columns)
```

```
plt.yticks(range(3), correlation.columns)
plt.colorbar(label="Correlation Coefficient")

for i in range(3):
for j in range(3):
plt.text(j, i, f"{correlation.iloc[i, j]:.2f}",
ha="center", va="center")

plt.title("Telemetry Parameter Correlation Heatmap")
plt.tight_layout()
plt.show()
```

8. Dashboard summary

```
print("\n" + "=" * 60)
print(" IoE TELEMETRY MONITORING DASHBOARD")
print("=" * 60)
print("Total Records :", len(database))
print("Number of Devices :", database["device_id"].nunique())
print("Average Temperature :",
round(database["temperature"].mean(), 2), "°C")
print("Average Humidity :",
round(database["humidity"].mean(), 2), "%")
print("Average Battery Level :",
round(database["battery"].mean(), 2), "%")
print("Normal Readings :",
int((database["status"] == "Normal").sum()))
print("Warning Readings :",
int((database["status"] == "Warning").sum()))
print("=" * 60)
```

9. Export datasets for visualization tools

```
database.to_csv("telemetry_data_for_visualization.csv", index=False)
summary.to_csv("telemetry_summary.csv")
device_summary.to_csv("device_wise_analytics.csv")

print("\nCSV files generated successfully:")
print("1. telemetry_data_for_visualization.csv")
print("2. telemetry_summary.csv")
print("3. device_wise_analytics.csv")

print("\nSimulation completed successfully.")
print("Import the CSV data into Power BI or Tableau.")
print("For Grafana, configure a compatible data source.")
```

10. Download the main dataset

files.download("telemetry_data_for_visualization.csv")

OUTPUT

```
======================================================================
       IoE TELEMETRY DATA INGESTION AND ANALYTICS
======================================================================

1. TELEMETRY DATA GENERATION
Data generated successfully.
Total records: 100
Total devices: 3

Sample Telemetry Records:
```

	timestamp	device_id	temperature	humidity	battery	status
0	2026-10-29 08:00:00	Device_03	29.75	57.40	62.51	Normal
1	2026-10-29 08:01:00	Device_01	30.66	58.35	77.46	Normal
2	2026-10-29 08:02:00	Device_03	30.68	48.48	85.73	Normal
3	2026-10-29 08:03:00	Device_03	30.26	69.53	83.01	Normal
4	2026-10-29 08:04:00	Device_01	27.38	70.40	64.00	Normal
5	2026-10-29 08:05:00	Device_01	26.13	53.06	97.74	Normal
6	2026-10-29 08:06:00	Device_03	23.48	64.94	86.89	Normal
7	2026-10-29 08:07:00	Device_02	31.30	69.74	77.72	Normal
8	2026-10-29 08:08:00	Device_03	27.47	61.81	80.59	Normal
9	2026-10-29 08:09:00	Device_03	26.77	66.78	70.98	Normal

2. TIME-SERIES DATA INGESTION
Data ingestion completed successfully.
Records stored: 100
Storage file: telemetry_database.csv

	timestamp	device_id	temperature	humidity	battery	status
0	2026-10-29 08:00:00	Device_03	29.75	57.40	62.51	Normal
1	2026-10-29 08:01:00	Device_01	30.66	58.35	77.46	Normal
2	2026-10-29 08:02:00	Device_03	30.68	48.48	85.73	Normal
3	2026-10-29 08:03:00	Device_03	30.26	69.53	83.01	Normal
4	2026-10-29 08:04:00	Device_01	27.38	70.40	64.00	Normal

3. TELEMETRY STATISTICAL ANALYSIS
Mean, minimum, and maximum values:

	temperature	humidity	battery
mean	28.27	61.42	75.04
min	20.03	45.69	50.54
max	38.12	81.88	99.53

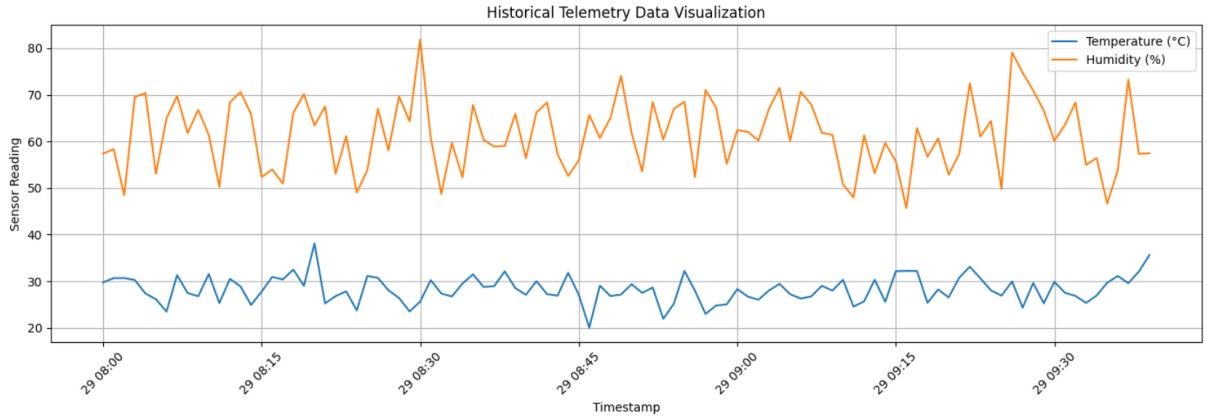
Device Status Summary:

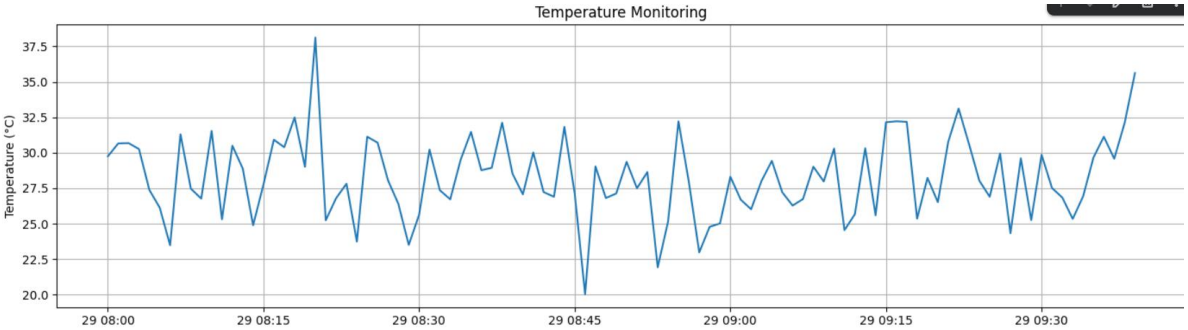
Device Status Summary:

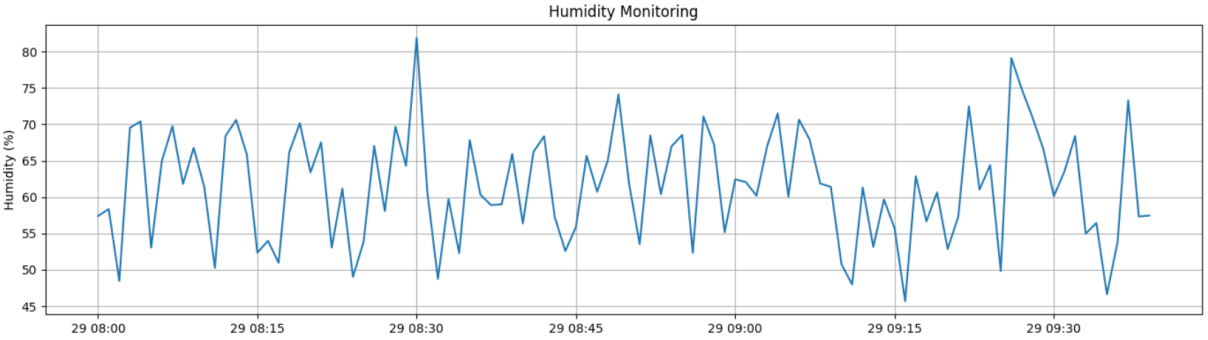
status	Number of Readings
Normal	78
Warning	22

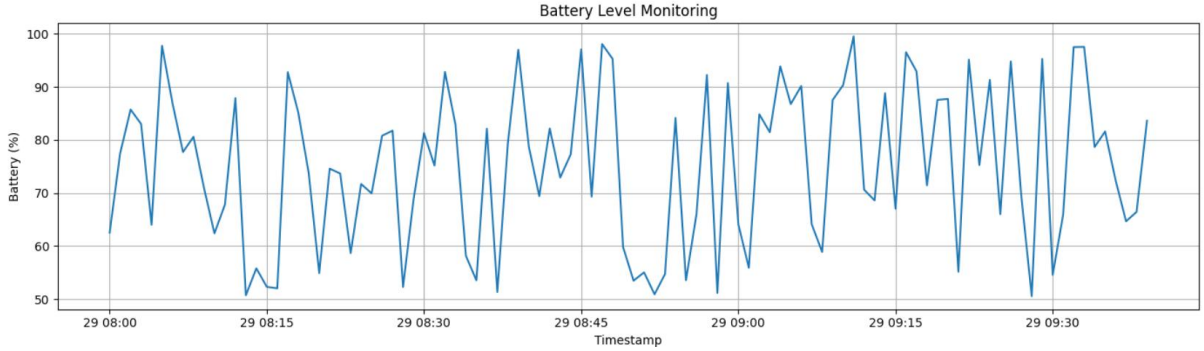
Device-Wise Analytics:

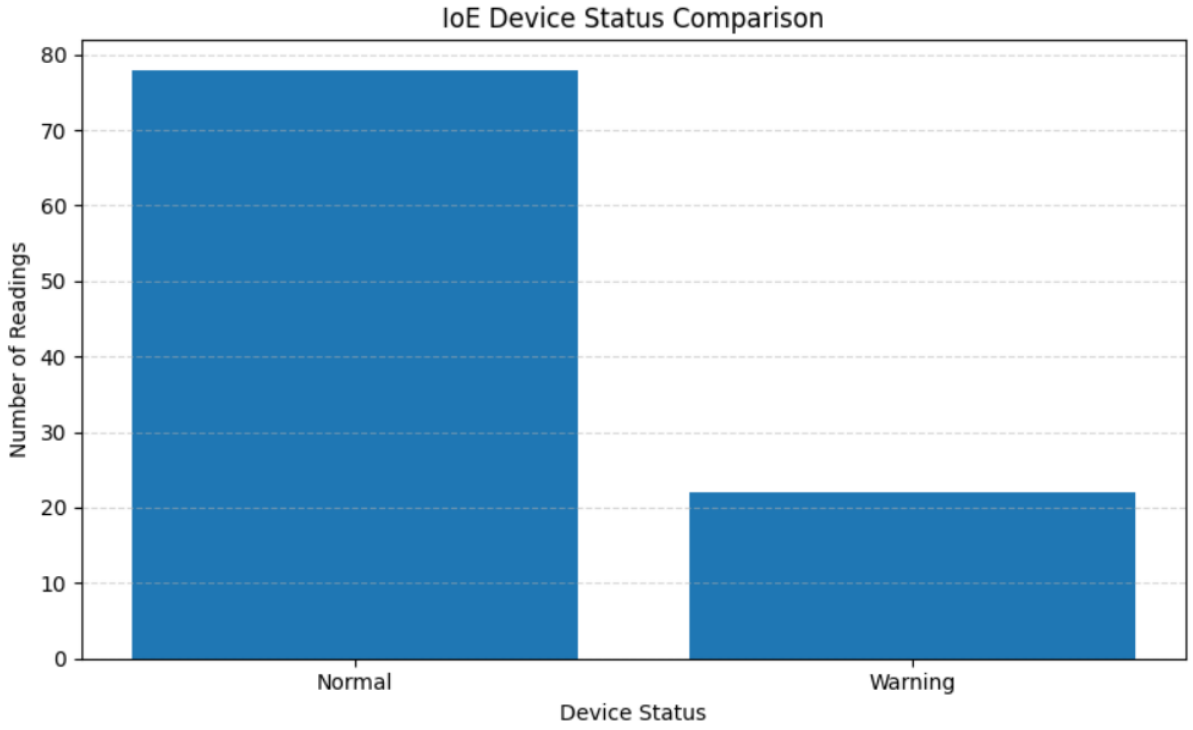
device_id	Average_Temperature	Average_Humidity	Average_Battery	Total_Readings
Device_01	28.61	60.17	74.86	33
Device_02	28.09	62.51	75.21	36
Device_03	28.12	61.48	75.05	31


Historical Telemetry Data Visualization
Temperature (°C)
Humidity (%)
Sensor Reading
Timestamp
29 08:00
29 08:15
29 08:30
29 08:45
29 09:00
29 09:15
29 09:30

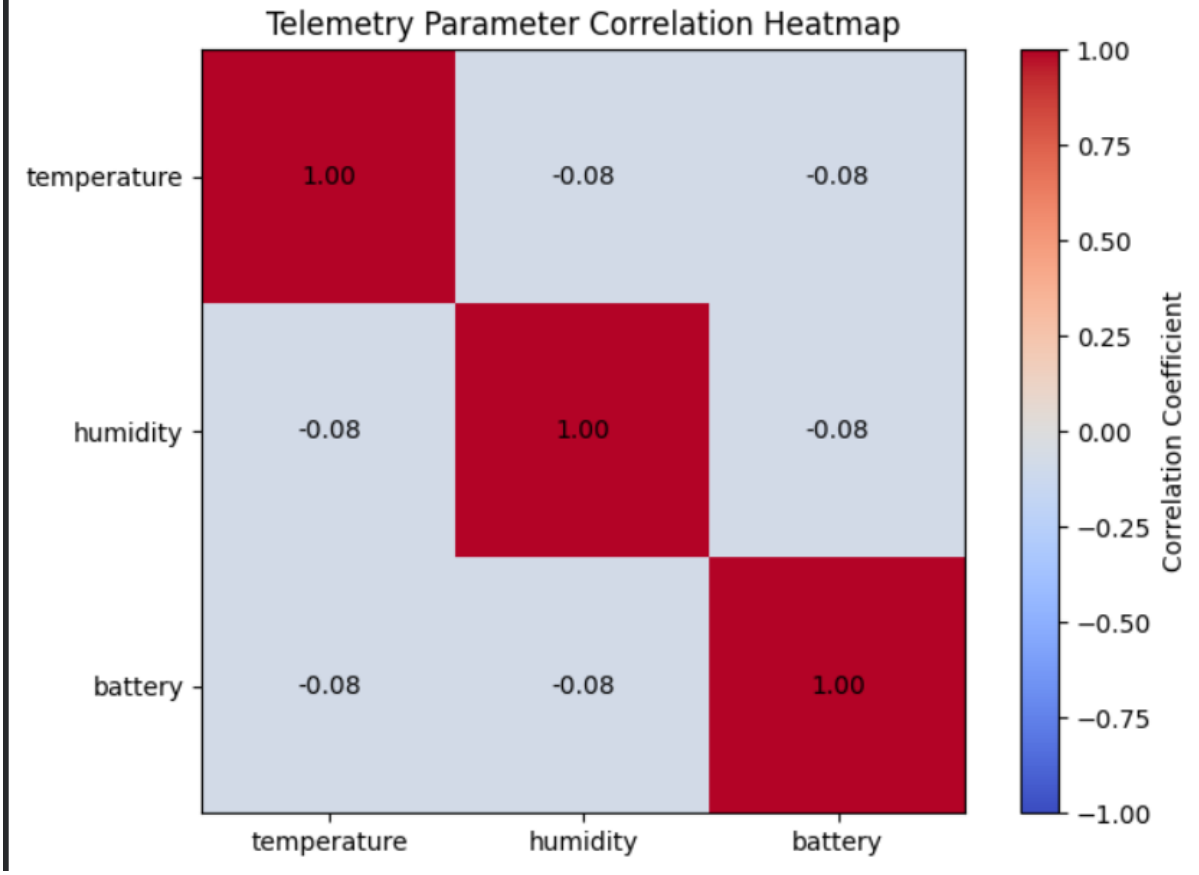

Temperature Monitoring
Temperature (°C)
29 08:00
29 08:15
29 08:30
29 08:45
29 09:00
29 09:15
29 09:30


Humidity Monitoring
Humidity (%)
29 08:00
29 08:15
29 08:30
29 08:45
29 09:00
29 09:15
29 09:30


Battery Level Monitoring
Battery (%)
Timestamp
29 08:00
29 08:15
29 08:30
29 08:45
29 09:00
29 09:15
29 09:30



Normal	78
Warning	22



```
============================================================
Total Telemetry Records : 100
Number of Devices       : 3
Average Temperature     : 28.27 °C
Average Humidity        : 61.42 %
Average Battery Level   : 75.04 %
Normal Readings         : 78
Warning Readings        : 22
============================================================
Dashboard Summary Generated Successfully.

9. DATA EXPORT COMPLETED
1. telemetry_data_for_visualization.csv
2. telemetry_summary.csv
3. device_wise_analytics.csv

The CSV files can be imported into:
Grafana - using a configured data source
Power BI - using Get Data > Text/CSV
Tableau - using Connect > Text File

All telemetry processing and visualizations completed successfully.
```